# MAKERERE UNIVERSITY

## College of Computing and Information Sciences

## School of Computing and Informatics Technology

## Department of Networks

### BSE4104 - Emerging Trends in Software Engineering

### AI Native and Agentic Engineering Capstone Project

### GROUP J EVENING

# WEEK 1: PROBLEM FRAMING AND AI NATIVE REQUIREMENTS REPORT

## SME PROCUREMENT PREPARATION AGENT


| NAMES | REGISTRATION NUMBERS |
|---|---|
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Work completed against weekly objectives.

- Selected and confirmed the SME Procurement Preparation Agent as the project use case.
- Defined a single primary end to end workflow from low-stock detection to reorder quantity recommendation to quotation comparison to draft purchase requisition finally to human approval routing.
- Completed the project charter covering problem statement, target users, AI value proposition, scope, assumptions, constraints, success measures and the minimum proposal statement.
- Developed the testable user stories with detailed acceptance criteria.
- Created a comprehensive AI Boundary Matrix clearly separating AI responsibilities, deterministic software ownership and mandatory human approval points.
- Designed the context diagram using draw.io.
- Set up the GitHub repository following the recommended structure.

Repository

GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent

Key engineering decisions and why.

| DECISIONS | WHY |
|---|---|
| Chose SME Procurement Preparation Agent. | Its well bounded, uses only synthetic data and has strong human requirements. |
| Kept numerical calculations deterministic. | Prevents hallucination of stock levels and prices which aligns with the course emphasis on justified AI use. |
| Mandatory human approval gate before any requisition leaves 'Draft' status. | Satisfies the safety boundary of no financial commitment or supplier selection can occur autonomously. |
| Designed for synthetic data only. | Ensures the team can fully control, test and demonstrate the system without depending on external approvals or confidential data. |
| Started with a simple approach. | Allows the team to understand the agent loop deeply before introducing heavier frameworks. |

Challenges encountered and current response.

- Balancing sufficient detail in user stories without over scoping. We responded by producing well focused stories which will prioritize implementation while keeping the stretch goals.
- Making the AI boundary matrix precise enough. We solved by iterating the matrix twice to clearly separate ‘AI may suggest’ from ‘AI may never decide.’
- Initial uncertainty about exact synthetic data volume. We agreed on a manageable starting set to be created in Week 2.

Plan for Week 2

Week 2 focus: Foundation-Model Engineering and Prompting

- Select and document an accessible foundation model.
- Integrate the model into a minimal working application.
- Create at least 10 test cases and record expected vs actual behaviour.
- Begin creation of the synthetic inventory and quotation datasets.
- Owners to be assigned in ClickUp .